THE GROUPOID ATTACHED TO A SMOOTH PRESENTATION IS SMOOTH AND THE CANONICAL MORPHISM IS AN EQUIVALENCE

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Editorial note. The covering-existence step is 'clear' from the surjective-smooth definition of f; carried as an explicit obligation. The transfer inputs are not carded and are declared external.

Allowed standard prerequisites (editorial).

- Presentation machinery. lemma-map-space-into-stack (carded L26690). Supplies the groupoid (U, R, s, t, c) and the fully faithful f_can.
- Property invariance under equivalence. lemma-property-morphism-equivalent (same chapter, NOT carded, declared external). Gives smoothness of s, t via the equivalences S_U -> U, S_R -> R.
- Presheaf transfer of smooth. lemma-map-presheaves-representable-by-spaces-transformation-property (same chapter, NOT carded, declared external). Reads s, t as images of the smooth f.
- Essential surjectivity criterion. Stacks, lemma characterize-essentially-surjective-when-ff (cross-chapter, named). Fully faithful plus locally essential surjectivity gives essential surjectivity.

Source reference. Original lemma 045A: sources/stacks-algebraic/pinned-original-source.tex, source lines 1118–1118. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04T4: sources/stacks-algebraic/pinned-original-source.tex, source lines 2250–2250. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0459: sources/stacks-algebraic/pinned-original-source.tex, source lines 1090–1090. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 046N: focused-seeds/geometry/S9-alg-stack-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X}$ be an algebraic stack over S. Let U be an algebraic space over S. Let $f : \mathcal{S}_U \to \mathcal{X}$ be a surjective smooth morphism. Let (U, R, s, t, c) be the groupoid in algebraic spaces and $f_{can} : [U/R] \to \mathcal{X}$ be the result of applying Lemma 04T4 to U and f. Then*

(1) *the morphisms s, t are smooth, and*
(2) *the 1-morphism $f_{can} : [U/R] \to \mathcal{X}$ is an equivalence.*

Proof. The morphisms s, t are smooth by Lemmas 0459 and 045A. As the 1-morphism f is smooth and surjective it is clear that given any scheme T and any object $a \in \mathrm{Ob}(\mathcal{X}_T)$ there exists a smooth and surjective morphism $T' \to T$ such that $a|'_T$ comes from an object of $[U/R]_{T'}$. Since $f_{can} : [U/R] \to \mathcal{X}$ is fully faithful, we deduce that $[U/R] \to \mathcal{X}$ is essentially surjective as descent data on objects are effective on both sides, see Stacks, Lemma 046N. □